✅ STEP 1 — Locked Context Confirmation

Check	Status	Evidence
FINAL-DESC-CMCMIS visible?	✓ **YES — fully loaded**	All 42 pages in context
Document version	v1.0 — FINAL — LOCKED	Sealed May 16, 2026
Project Code	CMCMIS_SIMPLIFIED	Confirmed
Phases sealed	0, 1, 2	Confirmed
Decisions locked	20 master + 6 stack + 5 confirmations = **31 locks**	All ✓ LOCKED
Current Phase	**Phase 3 — Database Design**	Today
Authority	This document supersedes all interim drafts	Confirmed

Locked anchors I will treat as non-negotiable today:

Anchor	Locked Value
Target table count	~65–75 (13 entity clusters)
Existing tables	~64 (yours, with data, never dropped in MVP)
DB engine	MySQL 8.x · InnoDB · utf8mb4
Driver / pattern	mysql2/promise + Repository Pattern (no ORM) (D2)
Bootstrap	≥ 2 Super Admin employee IDs via env CSV (D11, BR-AUTH-05)
New equipment default	PENDING_VERIFICATION (D10, BR-EQP-10)
Roles	5 only — Super Admin, Lab In-Charge, Lab Engineer, Normal, View-Only (C1)
RBAC model	User → Role → Permission → Resource:Action (3 layers, BR-RBAC-03)
MVP-critical clusters	1 (Identity), 3 (Equipment), 4 (Job Lifecycle), 10 (Audit), 12 (Lookups)

all are okay and my answer as yes you can proceed with this.

```
PRIORITY 1 — TODAY/THIS WEEK
════════════════════════════
Cluster 1   Identity & Access     (8–10 tables)
    ↓
Cluster 3   Equipment Master      (8–10 tables)
    ↓
Cluster 4   Job Lifecycle         (6–8 tables)
    ↓
Cluster 10  Audit & Logs          (3–4 tables)
    ↓
Cluster 12  Lookups               (5–7 tables)
────────────────────────────
TOTAL MVP-critical: ~30–39 tables

PRIORITY 2 — SKETCH ONLY (do not finalise)
══════════════════════════════════════════
Cluster 5   Calibration           (6–8)
Cluster 6   Maintenance           (4–5)
Cluster 2   Organisation          (4–5)
Cluster 9   Documents             (3–4)

PRIORITY 3 — DEFER (Phase 2)
════════════════════════════
Cluster 7   Scheduling
Cluster 8   Procurement
Cluster 11  Notifications
Cluster 13  Reporting
```

we have to design entire database with all the cluster and all the SCHEMAS today it self not in future.
I will provide you all the instructions for the same carefully and you have to work accordingly. And give me back responses in my style.

⚠️ STEP 7 — Things I'll Flag Loudly As I Audit

I'll surface these immediately, not bury them:

Flag	Meaning	Example
🔴 **BR-VIOLATION**	Existing table contradicts a locked business rule	Equipment table has no `registered_by` → violates BR-EQP-09
🟡 **DECISION-NEEDED**	Genuine fork; I can't pick without you	Two tables both look like `users` — which is canonical?
🟢 **STRAIGHTFORWARD**	Clean keep/modify, no debate	`tbl_division` → maps cleanly to Cluster 2
🟣 **ORPHAN-CANDIDATE**	Likely dead but I need your O2 classification to confirm	Table with 0 rows or last write in 2019
⚪ **REVISION-CANDIDATE**	Forces a v1.1 to the FINAL-DESC document	Existing schema makes a locked decision impractical

<u>REFER DOCUMENT JORDAR MAST KAY NO GHATE THE POWER OF AI AND THE ULTIMATE POWER OF Claude.ai</u>

Existing DB schema dump (~64 tables) ⏳ ======please refer attached files Super Admin employee IDs ⏳ ===== SA79900 AND AC77777 Table classification: in-use / dead / unsure ⏳ ===== I DON'T KNOW PLEASE FIND AND TEMM ME WITH YOUR LOGIC AND OUR FINAL DOCUMENT AND LOCKED STATE

please keep in mind above attached .sql and .csv files till deployment and now the hardest and the thought work do by yourself. here i don't know what table schemas do we needed as attached above and what are the fully matches with our requirement and what are un useless i don't know please put your all energy to retrieve all this things. design all the things related to db and ask me questions when ever needed at the end of this session we have to complete our entire database design please keep all the schemas and for all the modules

specially for auth module tell me all the new schemas requirement keep in my mind also compare new and old table created schemas by you. tell me all the things.

we have to design for the all the modules as=====> attached image for the same refer it

TAKE YOUR TIME KEEP ENERGY LEVEL UP DESIGN LIKE DATABASE ENGINEER. NEW OLD UN USED USED HELPFULL ALL THE THINGS.

REFER DOCUMENT JORDAR MAST KAY NO GHATE THE POWER OF AI AND THE ULTIMATE POWER OF Claude.ai

FINAL_MVP_DB.md open in Antigravity when ever you want to see it.

please refer attached document for the MVP as minimal viable product for the same. Now onwards please refer for it as database design and I want that please locked with it. And please locked it with your grateful and very intelligence mind and mindset.

please refer that document and below I provide you all the 8 Question answers and my THOUGHTS for the database design.

Q1 (🔴 BLOCKING for Cluster 3) — Where is `cmms_cont_mst`?

==============➔ please refer it as cmms_cont_mst like as I GIVE YOU AND I AGREED WITH YOU FOR THIS as I give you permissions to create it new table's schema because I don't have that table in my database as of now.

Q2 (🟡 Cluster 1) — Are SA79900 and AC77777 existing employees?**==============➔not exist as of now. Please seed or insert it into the database.**

Q3 (🟡 Cluster 1) — Migration of 565 existing user-role rows

===============➔ for this `cmms_userrole_mst` as of now please skip migration I don’t want any migration right now but we have to do this kind of thing as Entire new with the ENTIRE NEW AUTH RBAC PERMISSIONS AND USER_ROLES and super admin as SA79900 or AC77777 have rights to assign role to the normal users and after than store assigned role and next time user logged in with that role as RBAC and with PERMISSIONS and keep in my we have only 5 roles in entire CMCMIS-SIMPLIFIED.

Q4 (🟡 Cluster 1) — Mapping 23 old roles → 5 new roles

===============➔ we have only 5 roles in entire CMCMIS-SIMPLIFIED.

Q5 (🟢 Cluster 3) — Can `cmms_division_hist` serve as equipment_status_history?

===============➔ keep it as separated, as of now please design it table’s schemas and locked it in.

Q6 (🟢 Cluster 4) — Should we add `job_request_status_history`?

===============➔ keep it as separated, as of now please design it table’s schemas and locked it in.

Q7 (🟡 Cluster 12) — Password policy strength?

===============➔in password policy please refer this all the things-----------in future SSO time we will do with the another context but as of now please keep in mind we have to keep fixed 7 letters password in which we have compulsory 2 capital alphabets upper case and 5 fixed digits.no any small caps no any small alphabets lower case no any special chars characters no any special symbols. Password is lifetime thing so there is no any expiry and there is no any password history field.

Q8 (🟢 Org context) — `cmms_emp_mst.EMM_DEPT` semantics

===============➔ I think please design new table's schema for it as of now design it new table.

please follow this also ===============================
=====➔Table name as in small caps and meaning full name as with _ and . to read it and interpret it from it's name.

here we are going to develop entire new AUTH RBAC 3-LAYES AS PERMISSIONS AND user roles so please put your all the and highest efforts to

design it with meaning full terms like as new completely auth stack and above all the things for this please refer and remember attached document.

there is only 5 roles in our entire system as cmcmis-simplified.
as of now password is like only as of now in future password given by SSO. But as of now refer password as employee-id and password are same.
also 2 new entries in cmmc_emp_mst as super admin ids.
T&ME and F&PE are two different section under TIMCD department. Please remember this also.
and at last please mention in document where and what kind of data is needed for the migration purpose.

***** instructions *****

Please put your extra and all efforts to design it entire database please think as senior database designer for an enterprise level.
I want do you best with your intelligence mind and min set.
please locked this attached file for the database related things please completely remember and lock it in your grate things and brain.
please give me response back in my style and in .md document end to end step by step with the entire database

related all the things ALL THE UPDATED VERSIONS AND ANSWERS AND QUESTIONS things and all the rest of the things in phase 3 please put your extremely HIGHER AND HIGHEST COMPLEX EFFORTS.

V 2.0

M1

For SA79900/AC77777 we need a valid SM_ID.

- **Choose one approach (DS decides — see Migration Data Requirements §18):**

Answer ===========➔with the new section given as below data seeding type.

Option -→Insert a brand-new "ADMIN" row into cmms_section_mst before

the SA79900/AC77777 inserts. Suggested values:

INSERT INTO cmms_section_mst

(SM_ID, SM_SHORTNAME, SM_NAME, SM_HEAD_NAME, SM_STATE,

SM_CREATED_BY, SM_CREATED_ON, SM_UPDATED_BY, SM_UPDATED_ON)

VALUES (9999, 'ADMIN', 'System Administration', NULL, 1,

'BOOTSTRAP', NOW(6), 'BOOTSTRAP', NOW(6));

MY RECOMMENDATION: Option B. Single new row, clearly labelled, doesn't

muddy the meaning of an existing section.

M2

Full name + designation + email for SA79900 and AC77777

Answer ===========→take as your own I only provide you three thins as IDS PASSWORDS FOR THE SM_ID FOLLOW ABBOVE M1 PART AS NER ENTIRE SECTION SEEDING. **SA79900 and AC77777 with the password as same as employee_id as Super Admin IDs. SA79900 and AC77777. System super admin as name and also describe by your self other things you have all the rights and I am fully agreed with you.**

M3

At least one vendor row in new `cmms_cont_mst` so the 5,704 `cmms_eqip_mst` rows with `EQM_MFRID` have valid FK targets

Answer ===========→ STRATEGY: Derive a placeholder vendor list from existing equipment data.

```
PROBLEM:   cmms_eqip_mst has 5,704 rows with EQM_MFRID values pointing at
           cmms_cont_mst — but cmms_cont_mst was missing from your DB.
           Now we've CREATED cmms_cont_mst but it's empty.
           The FK constraints on the 5,704 rows are technically orphaned
           (the constraint is "deferred" since cmms_cont_mst row 0 didn't exist).

STRATEGY:  Derive a placeholder vendor list from existing equipment data.

SQL TO RUN (gives you a list to bulk-INSERT into cmms_cont_mst):

  SELECT DISTINCT
     EQM_MFRID                                              AS proposed_CMM_CONT_ID,
     COALESCE(EQM_MFG_MODEL_NAME, CONCAT('Vendor #', EQM_MFRID))
                                                            AS proposed_CMM_CONT_NAME,
     'MFR'                                                  AS proposed_CMM_CONT_TYPE
    FROM cmms_eqip_mst
   WHERE EQM_MFRID IS NOT NULL
   ORDER BY EQM_MFRID;

Then for each distinct EQM_MFRID:
  INSERT INTO cmms_cont_mst (CMM_CONT_ID, CMM_CONT_NAME, CMM_CONT_TYPE,
                             CMM_CONT_STATE_FLAG, CMM_CONT_CREATED_BY,
                             CMM_CONT_UPDATED_BY)
  VALUES (?, ?, 'MFR', 1, 'BOOTSTRAP', 'BOOTSTRAP');
```

SQL TO RUN (gives you a list to bulk-INSERT into cmms_cont_mst):

SELECT DISTINCT

EQM_MFRID AS proposed_CMM_CONT_ID,

COALESCE(EQM_MFG_MODEL_NAME, CONCAT('Vendor #', EQM_MFRID))

AS proposed_CMM_CONT_NAME,

'MFR' AS proposed_CMM_CONT_TYPE

FROM cmms_eqip_mst

WHERE EQM_MFRID IS NOT NULL

ORDER BY EQM_MFRID;

Then for each distinct EQM_MFRID:

INSERT INTO cmms_cont_mst (CMM_CONT_ID, CMM_CONT_NAME, CMM_CONT_TYPE,

CMM_CONT_STATE_FLAG, CMM_CONT_CREATED_BY,

CMM_CONT_UPDATED_BY)

VALUES (?, ?, 'MFR', 1, 'BOOTSTRAP', 'BOOTSTRAP');

M4 as follow M3

DIECT ANS.
M5 If unmapped, legacy data simply doesn't appear in new section-filtered views. Acceptable for MVP.

M6 Default: Do not auto-create**. Super Admin onboards on demand. Faster bootstrap, cleaner audit.**

M7 Default: All existing JCs →`VERIFIED_CLOSED` (assume historical jobs are closed).

M8 Default: If `JR_SECTIONJOB_NO IS NOT NULL` → `ASSIGNED`; else → `SUBMITTED`

M9 NULL on bootstrap; set later by Super Admin

M10 Already isolated; no MVP impact
M11 in prod, 10 in dev/test

M12 Locked defaults YES ession timeout preference (default JWT 15min + refresh 7 days per D17)

Prompt to Claude.ai

FIRST OF ALL PLEASE LOCKED 🔒 IN VERSION 2.0 AS FINAL DOCUMENT as green flags. i agreed with the all the things please complete this database design and seeding files generation phase as phase 3 now.

for the all the questions as M1 to M12 follow this

M1 For SA79900/AC77777 we need a valid SM_ID. • Choose one approach (DS decides — see Migration Data Requirements §18): Answer ===========èwith the new section given as below data seeding type.

Option ---- Insert a brand-new "ADMIN" row into cmms_section_mst before the SA79900/AC77777 inserts. Suggested values: INSERT INTO cmms_section_mst (SM_ID, SM_SHORTNAME, SM_NAME, SM_HEAD_NAME, SM_STATE, SM_CREATED_BY, SM_CREATED_ON, SM_UPDATED_BY, SM_UPDATED_ON) VALUES (9999, 'ADMIN', 'System Administration', NULL, 1, 'BOOTSTRAP', NOW(6), 'BOOTSTRAP', NOW(6));

M2 Full name + designation + email for SA79900 and AC77777 Answer ===========ètake as your own I only provide you three thins as IDS PASSWORDS FOR THE SM_ID FOLLOW ABBOVE M1 PART AS NER ENTIRE SECTION SEEDING. SA79900 and AC77777 with the password as same as employee_id as Super Admin IDs. SA79900 and AC77777. System super admin as name and also describe by your self other things you have all the rights and I am fully agreed with you.

M3 At least one vendor row in new cmms_cont_mst so the 5,704 cmms_eqip_mst rows with EQM_MFRID have valid FK targets Answer ===========è STRATEGY: Derive a placeholder vendor list from existing equipment data.

SQL TO RUN (gives you a list to bulk-INSERT into cmms_cont_mst):

SELECT DISTINCT

EQM_MFRID AS proposed_CMM_CONT_ID,

```
   COALESCE(EQM_MFG_MODEL_NAME, CONCAT('Vendor #', EQM_MFRID))
                                            AS proposed_CMM_CONT_NAME,
   'MFR'                                        AS proposed_CMM_CONT_TYPE
  FROM cmms_eqip_mst
 WHERE EQM_MFRID IS NOT NULL
 ORDER BY EQM_MFRID;
```

Then for each distinct EQM_MFRID: INSERT INTO cmms_cont_mst (CMM_CONT_ID, CMM_CONT_NAME, CMM_CONT_TYPE, CMM_CONT_STATE_FLAG, CMM_CONT_CREATED_BY, CMM_CONT_UPDATED_BY) VALUES (?, ?, 'MFR', 1, 'BOOTSTRAP', 'BOOTSTRAP');

M4 as follow M3

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******* instructions ***** Please put your extra and all efforts to design it entire database please think as senior database designer for an enterprise level. I want do you best with your intelligence mind and min set. please locked this attached file for the database related things please completely remember and lock it in your grate things and brain. please give me response back in my style and in .md document end to end step by step with the entire database related all the things ALL THE UPDATED VERSIONS AND ANSWERS AND QUESTIONS things and all the rest of the things in phase 3 please put your extremely HIGHER AND HIGHEST COMPLEX EFFORTS.**

please provide me this all the things 3. I (Claude) write the 11 migration files | | 001..010 + 099, exactly as described in §17.1. | | | | 4. I write the migration runner (node script, idempotent). | | | | 5. DS provisions a local dev MySQL with the existing 64-table| | dump loaded, runs the migrations, verifies §17 checklist. | | | | 6. We test the full bootstrap path: | | - SA79900 logs in with password 'SA79900' → JWT issued | | - SA79900 creates a test user (e.g. 'DS00001') | | - DS00001 logs in with password 'DS00001' → JWT issued | | - DS00001 cannot access /admin/users (403) | | | | 7. Lock the seed scripts as v2.0 "RUNTIME READY". | | | | 8. Move to Phase 4 — first feature module wired end-to-end.

please complete phase 3.0 and provide me files as all the seeding files and codes and all the things and at last one document for this all.

RUNNING SEEDING

keep in mind I uses for the database as tools like entire XAMPP Control Panel with Apache mysql and phpMyAdmin.
so please refer this in entire project you have to work accordingly.
also I have entire database with around 396,000+ rows in phpMyAdmin, which is completely running properly.

please refer as of now db name as final because in organization database on server they placed name as final in phpMyAdmin. With the database collation as utf8_unicode_ci {utf8_unicode_ci} and type as InnoDB.

Also in my all the 64 tables I have this above mentioned collation please keep in mind this things.

```
SELECT user_id, employee_id, is_active, is_locked,
password_hash_set_at FROM users;
```

```
SELECT u.employee_id, r.role_code, ur.assigned_by, ur.assigned_at
  FROM user_roles ur
  JOIN users u ON u.user_id = ur.user_id
  JOIN roles r ON r.role_id = ur.role_id;
```

```
SELECT COUNT(*) AS super_admin_perm_count
  FROM role_permissions rp
  JOIN roles r ON r.role_id = rp.role_id
 WHERE r.role_code = 'SUPER_ADMIN';
```

```
SELECT d.department_code, s.section_code, s.equipment_category
  FROM sections s
  JOIN departments d ON d.department_id = s.department_id;
```

```
SELECT SM_ID, SM_SHORTNAME, SM_NAME FROM
cmms_section_mst WHERE SM_ID = 9999;
```

```sql
SELECT EMM_ID, EMM_NAME, EMM_DESIGNATION, EMM_DEPT
  FROM cmms_emp_mst WHERE EMM_ID IN ('SA79900','AC77777');
```

```sql
SELECT actor_employee_id, action, entity_type, entity_id,
occurred_at, notes
  FROM audit_log ORDER BY audit_id;
```

cmd

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm i

added 15 packages, and audited 16 packages in 4s

4 packages are looking for funding

run `npm fund` for details

found 0 vulnerabilities

```
E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm fund
cmcmis-phase3-migrations@2.0.0
├── https://dotenvx.com
│   └── dotenv@16.6.1
├── https://opencollective.com/express
│   └── iconv-lite@0.7.2
├── https://github.com/sponsors/wellwelwel
│   └── lru.min@1.1.4
└── https://github.com/mysqljs/sql-escaper?sponsor=1
    └── sql-escaper@1.3.3


E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm i


up to date, audited 16 packages in 1s


4 packages are looking for funding
  run `npm fund` for details


found 0 vulnerabilities
```

```
E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm list --depth=0

cmcmis-phase3-migrations@2.0.0 E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3

├── bcryptjs@2.4.3

├── dotenv@16.6.1

└── mysql2@3.22.3



E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>copy .env.example .env

Overwrite .env? (Yes/No/All): no

        0 file(s) copied.


E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate:dry


> cmcmis-phase3-migrations@2.0.0 migrate:dry

> node runner/run-migrations.js --dry-run


✗ FATAL: Access denied for user 'root'@'localhost' (using password: YES)


E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate:dry
```

> cmcmis-phase3-migrations@2.0.0 migrate:dry

> node runner/run-migrations.js --dry-run

✗ FATAL: Unknown collation: 'utf8mb4_0900_ai_ci'

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate:dry

> cmcmis-phase3-migrations@2.0.0 migrate:dry

> node runner/run-migrations.js --dry-run

✗ FATAL: COLLATION 'utf8_unicode_ci' is not valid for CHARACTER SET 'utf8mb4'

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate:dry

> cmcmis-phase3-migrations@2.0.0 migrate:dry

> node runner/run-migrations.js --dry-run

Migration runner starting (DRY-RUN)

  DB: root@localhost:3306/final

  Bcrypt rounds: 10

  Super Admin IDs: SA79900,AC77777

Migrations dir: E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3\migrations

Found 12 migration file(s).

[DRY] 001__create_new_tables.sql (would run, checksum=b5d623145b61…)

[DRY] 002__alter_legacy_tables.sql (would run, checksum=daf4a4ba1f28…)

[DRY] 003__pre_bootstrap_admin_section.sql (would run, checksum=79d0f15f96a9…)

[DRY] 004__seed_super_admin_employees.sql (would run, checksum=d4b786edbfe6…)

[DRY] 005__seed_roles.sql (would run, checksum=84bf9747da9f…)

[DRY] 006__seed_permissions.sql (would run, checksum=abff2356e456…)

[DRY] 007__seed_role_permissions.sql (would run, checksum=c0fa0387d4a6…)

[DRY] 008__seed_super_admin_users.js (would run, checksum=568653fc2f25…)

[DRY] 009__seed_org_departments_sections.sql (would run, checksum=bd5b090dfb7b…)

[DRY] 010__seed_lookups_and_audit.sql (would run, checksum=8e6cad89996a…)

[DRY] 050__backfill_cmms_cont_mst.sql (would run, checksum=63480802ca09…)

[DRY] 099__isolate_legacy_unused.sql (would run, checksum=8d48b6989b3d…)

✓ All migrations processed.

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate

> cmcmis-phase3-migrations@2.0.0 migrate

> node runner/run-migrations.js

Migration runner starting (APPLY)

DB: root@localhost:3306/final

Bcrypt rounds: 10

Super Admin IDs: SA79900,AC77777

Migrations dir: E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3\migrations

Found 12 migration file(s).

▶ 001__create_new_tables.sql

✓ 001__create_new_tables.sql (717ms)

▶ 002__alter_legacy_tables.sql

✗ 002__alter_legacy_tables.sql failed: You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'DELIMITER //

CREATE PROCEDURE `_cmcmis_safe_alter`(

IN p_kind VARCHAR(20)...' at line 1

✗ FATAL: You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'DELIMITER //

CREATE PROCEDURE `_cmcmis_safe_alter`(

IN p_kind VARCHAR(20)...' at line 1

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate:reset

> cmcmis-phase3-migrations@2.0.0 migrate:reset

> node runner/run-migrations.js --reset

⚠ --reset will DROP the schema_migrations table. (Tables themselves untouched.)

⚠ Re-running migrations afterwards will re-execute every file.

✓ schema_migrations dropped.

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate

> cmcmis-phase3-migrations@2.0.0 migrate

> node runner/run-migrations.js

```
Migration runner starting (APPLY)
  DB: root@localhost:3306/final
  Bcrypt rounds: 10
  Super Admin IDs: SA79900,AC77777
  Migrations dir: E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3\migrations
  Found 12 migration file(s).

▶ 001__create_new_tables.sql
✓ 001__create_new_tables.sql  (5ms)
▶ 002__alter_legacy_tables.sql
✓ 002__alter_legacy_tables.sql  (7545ms)
▶ 003__pre_bootstrap_admin_section.sql
✓ 003__pre_bootstrap_admin_section.sql  (9ms)
▶ 004__seed_super_admin_employees.sql
✓ 004__seed_super_admin_employees.sql  (17ms)
▶ 005__seed_roles.sql
✓ 005__seed_roles.sql  (7ms)
▶ 006__seed_permissions.sql
✓ 006__seed_permissions.sql  (8ms)
▶ 007__seed_role_permissions.sql
✓ 007__seed_role_permissions.sql  (29ms)
▶ 008__seed_super_admin_users.js
[008] bcrypt rounds = 10
```

✗ 008__seed_super_admin_users.js failed: Data too long for column 'created_by' at row 1

✗ FATAL: Data too long for column 'created_by' at row 1

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate

> cmcmis-phase3-migrations@2.0.0 migrate

> node runner/run-migrations.js

Migration runner starting (APPLY)

DB: root@localhost:3306/final

Bcrypt rounds: 10

Super Admin IDs: SA79900,AC77777

Migrations dir: E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3\migrations

Found 12 migration file(s).

[SKIP] 001__create_new_tables.sql (already applied, checksum matches)

[SKIP] 002__alter_legacy_tables.sql (already applied, checksum matches)

[SKIP] 003__pre_bootstrap_admin_section.sql (already applied, checksum matches)

[SKIP] 004__seed_super_admin_employees.sql (already applied, checksum matches)

[SKIP] 005__seed_roles.sql (already applied, checksum matches)

[SKIP] 006__seed_permissions.sql (already applied, checksum matches)

[SKIP] 007__seed_role_permissions.sql (already applied, checksum matches)

▶ 008__seed_super_admin_users.js

[008] bcrypt rounds = 10

[008] widening *_by columns to VARCHAR(20) (idempotent)…

[008]   [WIDENED] users.created_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] users.updated_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] user_roles.assigned_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] role_permissions.granted_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] departments.created_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] departments.updated_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] sections.created_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] sections.updated_by → VARCHAR(20) NULL DEFAULT NULL

[008]   [WIDENED] audit_log.actor_employee_id → VARCHAR(20) NOT NULL

[008]   [WIDENED] cmms_cont_mst.CMM_CONT_CREATED_BY → VARCHAR(20) NOT NULL

[008]   [WIDENED] cmms_cont_mst.CMM_CONT_UPDATED_BY → VARCHAR(20) NOT NULL

[008]   [INSERTED] user "SA79900" → SUPER_ADMIN

[008]   [INSERTED] user "AC77777" → SUPER_ADMIN

[008] ✓ complete. inserted=2, skipped=0

✓ 008__seed_super_admin_users.js (686ms)

▶ 009__seed_org_departments_sections.sql

✗ 009__seed_org_departments_sections.sql failed: Illegal mix of collations (utf8_unicode_ci,IMPLICIT) and (utf8_general_ci,IMPLICIT) for operation '='

✗ FATAL: Illegal mix of collations (utf8_unicode_ci,IMPLICIT) and (utf8_general_ci,IMPLICIT) for operation '='

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate

> cmcmis-phase3-migrations@2.0.0 migrate

> node runner/run-migrations.js

Migration runner starting (APPLY)

DB: root@localhost:3306/final

Bcrypt rounds: 10

Super Admin IDs: SA79900,AC77777

Migrations dir: E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3\migrations

Found 12 migration file(s).

[SKIP] 001__create_new_tables.sql (already applied, checksum matches)

[SKIP] 002__alter_legacy_tables.sql (already applied, checksum matches)

[SKIP] 003__pre_bootstrap_admin_section.sql (already applied, checksum matches)

[SKIP] 004__seed_super_admin_employees.sql (already applied, checksum matches)

[SKIP] 005__seed_roles.sql (already applied, checksum matches)

[SKIP] 006__seed_permissions.sql (already applied, checksum matches)

[SKIP] 007__seed_role_permissions.sql (already applied, checksum matches)

[SKIP] 008__seed_super_admin_users.js (already applied, checksum matches)

▶ 009__seed_org_departments_sections.sql

✓ 009__seed_org_departments_sections.sql (34ms)

▶ 010__seed_lookups_and_audit.sql

✓ 010__seed_lookups_and_audit.sql (11ms)

▶ 050__backfill_cmms_cont_mst.sql

✓ 050__backfill_cmms_cont_mst.sql (140ms)

▶ 099__isolate_legacy_unused.sql

✗ 099__isolate_legacy_unused.sql failed: You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'DELIMITER //

CREATE PROCEDURE `_cmcmis_safe_rename`(

IN p_old_name VARCHAR...' at line 1

✗ FATAL: You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'DELIMITER //

CREATE PROCEDURE `_cmcmis_safe_rename`(

IN p_old_name VARCHAR...' at line 1

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run migrate

> cmcmis-phase3-migrations@2.0.0 migrate

> node runner/run-migrations.js

Migration runner starting (APPLY)

DB: root@localhost:3306/final

Bcrypt rounds: 10

Super Admin IDs: SA79900,AC77777

Migrations dir: E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3\migrations

Found 12 migration file(s).

[SKIP] 001__create_new_tables.sql (already applied, checksum matches)

[SKIP] 002__alter_legacy_tables.sql (already applied, checksum matches)

[SKIP] 003__pre_bootstrap_admin_section.sql (already applied, checksum matches)

[SKIP] 004__seed_super_admin_employees.sql (already applied, checksum matches)

[SKIP] 005__seed_roles.sql (already applied, checksum matches)

[SKIP] 006__seed_permissions.sql (already applied, checksum matches)

[SKIP] 007__seed_role_permissions.sql (already applied, checksum matches)

[SKIP] 008__seed_super_admin_users.js (already applied, checksum matches)

[SKIP] 009__seed_org_departments_sections.sql (already applied, checksum matches)

[SKIP] 010__seed_lookups_and_audit.sql (already applied, checksum matches)

[SKIP] 050__backfill_cmms_cont_mst.sql (already applied, checksum matches)

▶ 099__isolate_legacy_unused.sql

✓ 099__isolate_legacy_unused.sql (299ms)

✓ All migrations processed.

POST-BOOTSTRAP VERIFICATION CHECKLIST (v2.0 §17)

✓ roles count == 5 → 5

✓ permissions count == 40 → 40

✓ role_permissions count > 100 → 122

✓ users count == 2 → 2

✓ user_roles count == 2 → 2

✓ departments count == 1 → 1

✓ sections count == 2 → 2

✓ cmms_emp_mst includes SA79900 → 1

✓ cmms_emp_mst includes AC77777 → 1

✓ cmms_section_mst includes ADMIN (9999) → 1

✓ lookup values seeded → 30

✓ audit_log has bootstrap rows → 12

✓ bcrypt: SA79900 password verifies as 'SA79900'

✓ bcrypt: AC77777 password verifies as 'AC77777'

✓ ALL 14 CHECKS PASSED — system is RUNTIME READY.

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>

E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE CODE\DATABASE\phase3>npm run test:bootstrap

> cmcmis-phase3-migrations@2.0.0 test:bootstrap

> node runner/test-bootstrap.js

Ignoring invalid configuration option passed to Connection: bcryptRounds. This is currently a warning, but in future versions of MySQL2, an error will be thrown if you pass an invalid configuration option to a Connection

═══ END-TO-END BOOTSTRAP TEST ═══

▶ Test 1: SA79900 logs in with password 'SA79900'

✓ SA79900 login successful

  role: SUPER_ADMIN

  permissions: 40 granted

▶ Test 2: SA79900 prepares DS00001 in cmms_emp_mst (legacy step), then creates user

✓ DS00001 created (user_id=3, role=NORMAL_USER, section_id=1)

▶ Test 3: DS00001 logs in with password 'DS00001'

✓ DS00001 login successful

  role: NORMAL_USER

  permissions: 13 granted (subset of Super Admin's 40)

▶ Test 4: DS00001 tries to create another user (should be FORBIDDEN)

✓ DS00001 correctly DENIED (no user:role-assign permission)

▶ Test 5: DS00001 attempts login with wrong password (should FAIL)

✓ Wrong password correctly rejected

failed_login_count incremented to: 1

▶ Test 6: Login with malformed password 'abc123' (should FAIL fast)

✓ Malformed password correctly rejected (before bcrypt)

▶ Test 7: Cleanup — reset DS00001 failed_login_count

✓ DS00001 state reset

═══ TEST SUMMARY ═══

✓ ALL TESTS PASSED — Phase 3 bootstrap is RUNTIME READY.

═══ Recent login_audit rows (last 8) ═══

(index)	employee_id	outcome	ip_address	attempt_at
0	'DS00001'	'FAILED_INVALID_FORMAT'	'127.0.0.1'	2026-05-17T11:45:25.974Z

```
| 1      | 'DS00001'   | 'FAILED_BAD_PASSWORD'   | '127.0.0.1' |
2026-05-17T11:45:25.970Z |
| 2      | 'DS00001'   | 'SUCCESS'               | '127.0.0.1' | 2026-05-
17T11:45:25.886Z |
| 3      | 'SA79900'   | 'SUCCESS'               | '127.0.0.1' | 2026-05-
17T11:45:25.698Z |
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E:\SOFTWAREs By DS\cmcmis-simplified\SOFTWARE
CODE\DATABASE\phase3>
```